

Languages of Geometry

Mathias Bernhard(✉)

Department of Architecture, ETH Zurich, Zurich, Switzerland
bernhard@arch.ethz.ch

Abstract. The history of computational design and CAD is often presented as inevitable and promoted as best fitting the designers' needs. A seemingly logical progression from ruler and compass to current software environments glosses over the fact that the modes of expression are rather limited. I refer to the constructive concepts behind a geometric thought model and production system as a language. Languages differ in how they express ideas or what ideas can be expressed at all, depending on the vocabulary and grammar in place. I present a collection of 22 geometric languages, a *Rosetta Form*, so to speak. Each language is presented as a standardized interactive web application, saying the geometric equivalent of "Hello, World!" first but highlighting the differences between them by modifying one parameter. The languages are presented as descriptions, images of the app, examples of their architectural relevance, and as part of a genealogy, a network of interrelations to overarching concepts.

Keywords: geometry · language · theory · history of computational design · interactive web applications

1 Introduction

Most articles on "the history of CAD" follow the simple storyline that one day, out of the blue, Ivan Sutherland invented all of CAD with *Sketchpad* [1]. Then, about every decade, this basis was extended first by the third dimension, then by semantic building information models (BIM), parameters, and finally, artificial intelligence (AI). This linear narrative is presented as progressive, inevitable, brand agnostic, and beneficial for architects and designers because it would follow their demands and fulfill their needs.

However, this is not the whole picture. Other CAD systems existed before, but economic, personal, and opportunity-related factors influenced the market dominance of some brands over others [2]. Alternative modeling concepts were invented in computer graphics [3], parametric design and semantic models are as old as CAD. Most importantly when planning to digitalize their work, architects were looking for tools that could produce drawings, not virtual twins of yet-to-be-built buildings [4]. This follows the assumption that the language of architects is 2D drawings consisting of lines and circular arcs.

Parallels between computational design and language have been established and described before, most notably related to procedural generation using spatial *grammars*

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[5–8], a concept borrowed from linguistics. They assume that in analogy to the syntactic tree diagram in phrase structure analysis, parts in a design can sequentially be replaced by more detailed ones from sets of parallel options, following strict rules to avoid combinatorial explosion and trim the solution space from non-sensical solutions.

This assumes that there is only one language of shapes and geometry – as if we would assume there is only English to communicate in written and spoken words [9]. Benjamin Lee Whorf famously said "Language shapes the way we think, and determines what we can think about". Now language, neither spoken nor geometric, is completely deterministic. There are things expressible in many different languages and Lera Boroditsky quotes William Shakespeare [10, 11] who has Juliet say "A rose by any other name would smell as sweet." This paper is about the geometric equivalent of the red flower, but it is called Rose, Trandafir, Méiguī, or Poce. At first sight, one might call all of them *rectangles*, but only one of them actually is, according to the strict meaning of the term. This both boring and artificial form serves as a case study. It is a form we are permanently surrounded by [12], and recognizing and distinguishing it is something apparently unique to humans [13].

Digital design, due to its origins in computer graphics and software development for different engineering disciplines, is firmly anchored in functional and technological aspects. However, architectural geometry affects everything, our culture, well-being and how we live as a society. The geometry of straight streets is a demonstration of power and an instrument for social control. Caroline Criado Perez [14] describes how the geometry of sanitary facilities equal by area means all but equality, severely affecting women's health. Alison B. Powell [15] illustrates the connection between the geometry of low-emission zones and pedestrian safety. The shapes we design are never geometry alone. *"A design cannot be disconnected from the values and assumptions in which it was created […]"* writes Ruben Pater [16] and the artist Peter Halley [17] declares a crisis in geometry, as even the most abstract of shapes in art is not a detached signifier alone, but always signifies something and evokes associations.

The objective of this research is to provide an overview and a comparative analysis, illustrate the languages' potential, show some of their blind spots, and present alternatives to cover them. It is not about claiming the superiority of one over the other. None of them is best for everything. Each excels at some tasks while requiring tremendous effort to succeed (if at all) at others.

2 Method

A common thing to do in languages, particularly programming languages, is Greeting to both begin a dialog and understand the most basic syntax. The famous print command "Hello, World!" is available in 592 languages on the platform *Rosetta Code* [18] – named after the famous tri-lingual *Rosetta Stone* – where common algorithms like sorting are available with code in different programming languages.

This research is somewhat the geometric equivalent, it presents the form shown in Fig. 1 in many different languages as geometric concepts, all implemented in the same programming language (JS). It could, therefore, be called *Rosetta Form.*

2.1 Language

This paper refers to "the thing shown in Fig. 1" as *form*. However, already the term chosen to describe the image implies a certain thought model and limits us from thinking differently about it. If we call it *shape*, which is both a noun and a verb, this refers to a series of physical, mostly subtractive steps to work out the form from an initial raw state in an operationalist manner. It would describe not so much the result itself but the sequence of steps necessary to get there, an important concept in, e.g., parametric modeling with history and constructive solid geometry (CSG) trees. If we call it an *object*, we make an unnecessary and limiting distinction between the red and the blue regions and classify it as a figure/ground logic. Alternatively, one could call it a *thing*, the most generic term and root of all things in an international effort of the all-encompassing ontology definition schema.org [19]. *Thing*, however, is not necessarily tangible, but it still refers to an isolated identifiable entity. If we call it *figure*, this would not only be confusing because it is the term used for graphical content in a publication but also has many anthropomorphic connotations and strong symbolic implications.

And finally, calling it a *rectangle* would be the most limiting term of them all. While the English term *rectangle* refers to right angles, the German *Viereck* refers to four corners, and the French *quadrilatère* to four sides. The term *rectangle* has even a political dimension. The term *rect* comes from the Latin *rectus*, meaning something (a pole) erected perpendicularly to the ground. It shares etymological roots with *reign*. Right (as in right angle) also opposes it to everything not right, wrong, or even left, right as in law and order, or right as in suitable, appropriate, genuine or real [20].

2.2 Solution Spaces

Each language presented here has a unique solution space. While they all intersect at – at least or maybe only – the location where the form in Fig. 1 is located, they all have different subspaces, some overlapping with one part of the other languages while being completely unreachable by others. The one modifiable parameter alters the form to results, geometric sentences, that cannot be expressed by others.

Two different approaches to exploring these spaces of geometric potential can be identified. One would be, "Given a language, what are the things that can be expressed?" The author has done this extensively for some of the languages [21–25]. Another one is, and this is where this paper wants to provide a basis for, "given a task, what is the best-suited language to talk about it, which one has the words (or degrees of freedom) necessary?".

3 Results

The research presented in this paper concerns the theoretical foundations of architectural geometry and digital design, and the results are, therefore, rather abstract. The primary result presented here is a collection consisting of 22 (currently, but the number keeps growing) "languages", geometric concepts or thought models for creating form. Each language has its own vocabulary, its own grammar and its very own range of stories to tell and ideas capable of expressing. A secondary result is an attempt at creating a genealogy about the relationships to common overarching themes exist between the languages.

3.1 Collection

The collection's elements are meticulously prepared in the same format. They can be presented on either a common overview web page[1] or individually on a dedicated page[2]. Each language in the collection is implemented in and visualized with an interactive web app written in p5js, showing a 500x500 pixel sky blue canvas with the identical "word/sentence" – or in geometrical terms: form, a 300 × 200 salmon-colored quadrilateral, rotated by 20° – at the center (see Fig. 1).

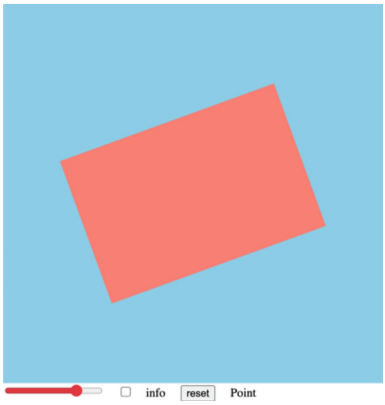


Fig. 1. The geometrical "Hello, World!" is expressed in all the collection's languages.

Below each canvas is a GUI containing a slider to modify one parameter (one dimension of the solution space), a checkbox to toggle between the rendered view and a debug view showing clues about the construction, a button to reset the initial state, and a label with the language's name. The appendix contains a detailed description of the construction behind each language, what parameter the slider modifies and how this affects the result, the language's relevance for architectural geometry and computational design, and to what concepts it is connected (see Sect. 3.2).

The languages' names in alphabetical order are Aggregation, Angle, Boolean, Cylinder, DistanceFunction, Epicycle, Fourier2D, Hyperbolic, LLM, NURBS, Offset, Parallelogram, Point, Pointcloud, Polar, Probability, Projection, Sphere, Superellipse, Tessellation, Tile, and Turtle. The Appendix contains a detailed description of each one of them. An illustrative triptych like the one exemplarily shown in Fig. 2 for every language can be consulted online[3].

The list is already long but not final. At the time of submitting this paper, a 23rd language based on cellular automata is being finalized, and many others (form-force, subdivision, complex numbers, etc.) are waiting in line. The goal evidently is not to be able to draw the exact same form repeatedly. Instead, isolating these concepts in an

[1] worbit.github.io/rosettaform.
[2] worbit.github.io/ < language-name > .
[3] worbit.github.io/illustrations.

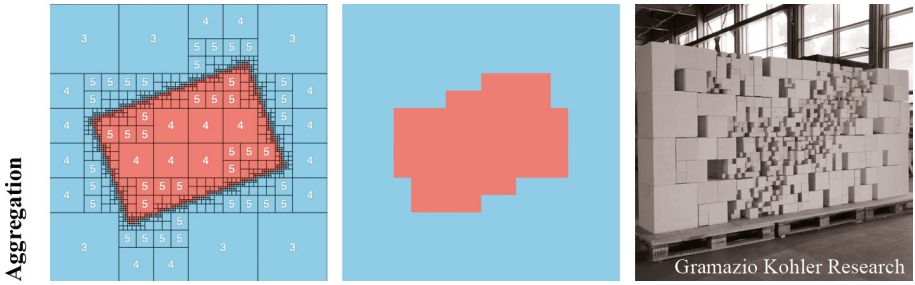


Fig. 2. Example triptych illustrating one language's behind-the-scenes look, a modified version, and an example of its architectural relevance.

interactive app makes them understandable in a teaching context, and starting from a location common to all solution spaces makes them comparable.

Almost anything can be expressed in almost every language, but some things become unnecessarily verbose. For example, one can approximate a rounded rectangle with *Points* of a polygon, but one needs more words, while e.g. as a *DistanceFunction*, one simply adds the radius to the formula.

3.2 Genealogy

The individual languages listed in the collection isolate one specific concept. Some can be considered a full-fledged system of expressions to build entire modeling environments. Others are more partial and must be combined with languages sharing similar concepts to cover a useful range of functions. It is thereby important to note that it is not a hierarchical classification tree, where each leaf (language node) is only connected to one branch, but all of them relate to at least two of these more general, overarching themes, resulting in the strongly connected graph shown in Fig. 3.

1. **Empty:** Basic geometric objects such as Points are placed in an empty space (Euclidean or other) with some coordinate system (Cartesian or other). These are then explicitly connected through lines and faces to a predefined topology.
2. **Field:** Space in any arbitrary number of dimensions is fully defined at every location as a scalar field, continuous or discrete. These values can be the result of a function or of physical measurements. Fields are the language of numerous environmental analyses and simulations, based on finite element analysis (FEA) or computational fluid dynamics (CFD).
3. **3D:** These languages' primary domain is three-dimensional space and the form shown here is a dimensionality reduction to 2D.
4. **Typology:** This family of languages refers to operations commonly used by architects and designers in conventional CAD environments to work on traditional typologies.
5. **Trigonometry:** Languages in this category are related to circular motion and rotation and require trigonometric functions for their construction.
6. **Uncertainty:** This class covers languages that bring a certain fuzziness, where absolute precision is impossible, just a form of statistically relevant close enough.

7. **Approximation:** These languages can handle selective resolutions and adaptive precision in approximating the result, which is important for the economy of space and time in computation.
8. **Fluid Topology:** With these languages, the relationship between entities (e.g. the connection of points with edges, the genus of forms, or the neighborhood relationships) is not predefined and static but dynamically computed and variable.

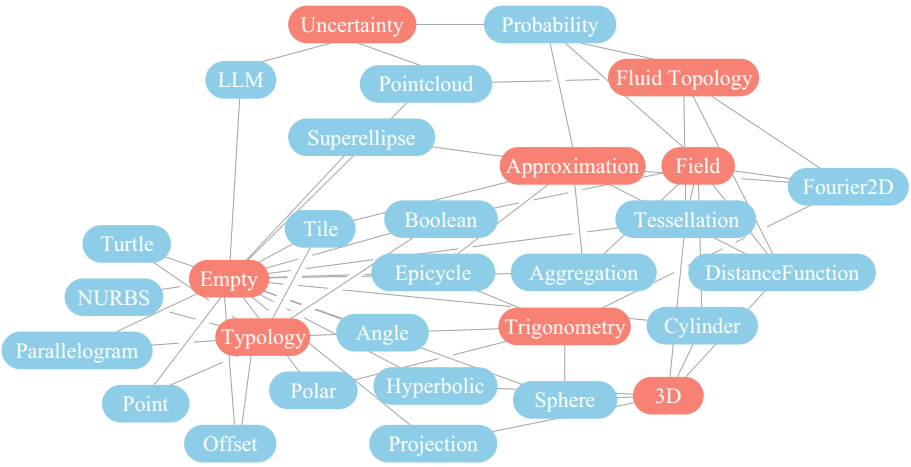


Fig. 3. Graph showing the genealogy of languages (blue) and overarching themes (red). (Color figure online)

4 Discussion

Many of today's challenges architects and designers face – resource scarcity, urbanization, climate change, to name a few – and the possible strategies – maintenance, repair, reuse – require new instruments that can handle the new vocabulary and new expressions of these novel tasks.

However, most of the tools we have are made for a different era, one focusing on growth, industrialization and standardization, breaking new ground, and real estate as a profitable investment. Most of the tools speak the languages related to Typology (see Sect. 3.2), allowing for the precise construction and fabrication of new, homogenous, crisp shapes placed in empty space on a white canvas, or the physically equivalent green lawn.

The geometric primitives available in conventional toolbars are ineligible to efficiently represent the heterogeneity and deviations from the ideal of partially battered and worn used building components. But instead of forcing the 3d scanned point cloud of an old and twisted wooden beam into the shape of a best-fitting cuboid – because that's the only language our tool understands – we need to ask what features and properties the language needs to speak about these uncertainties and approximations, and this paper can hopefully provide some answers.

Appendix Language Descriptions

The following pages contain a description of each language and the constructive concepts behind it, a paragraph on what parameter the slider modifies and how this deforms the result (a), the language's relevance (b), and to what concepts (c) the language relates. The languages are listed in alphabetical order.

1. **Aggregation**: The base language is a signed distance function (SDF) (see *DistanceFunction*), but instead of sampling space in a uniform, dense grid (raster), the function is evaluated sparsely with a quadtree. Squares are recursively subdivided into four smaller squares if the absolute distance value d at the center is smaller than half the square's diagonal, meaning the form's boundary passes through the respective square.
   a. The parameter controls the level of recursions, with lower numbers resulting in a coarse approximation, and higher numbers in a more accurate definition of the form with smaller leaf nodes.
   b. The assembly of a big form from small discrete elements is the core and essence of most architecture, from simple brick walls to highly elaborate mosaics. Unlike with Tiles, this aggregation type uses differently sized elements and thereby allows for an adaptive resolution [26]. Concepts like recursion and hierarchical data structure are also fundamental in computer science, and quadtree subdivision for rendering (z-buffer) is one of the oldest algorithms in computer graphics [27].
   c. Empty, Field, Approximation
2. **Angle**: This is, strictly speaking, the only real *rectangle* in the collection, as it is defined by constraining the interior angle α at three corners (the fourth being $2\pi - 3\alpha$), along with two edge lengths a and b. A form with only right angles having four corners is true only in the Euclidean plane. On a sphere, such an object can have three corners and more than four corners on a hyperbolic surface.
   a. Modifying the angle α turns the form into an arbitrary quadrilateral, with the fourth angle and lengths c and d calculated to close the shape.
   b. The ability to construct precise angles or regular subdivisions of the full circle with a compass and ruler is a fundamental and foundational principle of all geometry. The "right" angle ($90°$, or $\pi/2$) has a particularly prominent position, is the subject of Euclid's Elements' 4th postulate, and is inseparably linked to human – especially "man" [28] – made structures, distinguishes the artificial from the natural and is important for civilization and industrialization.
   c. Empty, Typology
3. **Boolean**: The shape is the Boolean intersection of two quarter discs. If they are rotated 180° relative to each other, the result is a rectangle.
   a. Rotating one of the quarter discs results in different intersection shapes, with a varying number of control points and both linear and arc connections.
   b. Combinations of basic primitives into more elaborate shapes through either Boolean union (or, ∪), intersection (and, ∩) or subtraction (not, –) or other operations are at the heart not only of any computational design environment (CSG) but also logic. With Boolean operations on boundary representation (BRep) objects, the necessary reconstruction of the topology can become difficult and prone to failure [29].

c. Empty, Typology, Field, Fluid Topology

4. **Cylinder**: The form is the cross section through a cylinder, initially having its axis in the cutting plane.
   a. The axis of the cylinder is rotated out of the cutting plane, until being perpendicular to it. In this case, the resulting cross-section is a circle. This language has a rectangular and a circular form in the same solution space.
   b. Two-dimensional sections through three-dimensional objects as a dimensionality reduction is at the heart of architectural representation and a very strongly codified language spoken (drawn) and interpreted by insiders. Convention has it, that a long narrow, rectangular shape stands for a cuboid object, where the section profile is invariant along the entire height. This assumption fails as soon as objects are more complex.
   c. Empty, Field, 3D
5. **DistanceFunction**: In the history of computer graphics, alternative methods to the most widespread BReps have always existed [30], sometimes referred to as function representation FRep [31], implicit or volumetric modeling [21, 23]. Instead of placing points in the void, they describe space as a scalar field, mainly as the result of signed distance functions (SDF) of the form $d = f(x,y,z)$, where d is the distance of point (x,y,z) to the object described by f with negative values being inside, positive values outside and the value 0 on the boundary of the object(s).
   a. The parameter p is added $(p > 0)$ to or subtracted $(p < 0)$ from the result of the SDF, eroding or dilating the form respectively by shifting the iso-level 0. If p is negative, the offset form naturally has rounded corners – something difficult to achieve with other languages.
   b. SDFs provide a very flexible and robust modeling experience with applications in architecture, design, medical or aerospace engineering [32], 3D-printing, and microstructure design [24]. They allow for smooth combinations, a compact and resolution-agnostic representation and effortless CSG operations, as forms are computed rather than constructed. Besides the creation of geometry, SDFs can serve as effective encoding for machine learning applications [33, 34] or collision control for robotics [35].
   c. Field, Fluid Typology, 3D
6. **Epicycle**: A series of points along the contour of the form is converted into complex numbers (x: real, y: imaginary) and transformed to the frequency domain using FFT where a low pass of size n filter is applied. The form is then created with a drawing machine, not unlike Dürer's spider and snake [36, 37], consisting of circles on top of circles with fixed rotation frequencies $\frac{1}{i}$, $i \in \{1, \ldots, n\}$, where the complex number specifies the radius (magnitude) and the phase shift (angle).
   a. Changing the size of the low pass filter n alters the accuracy of the final form. Higher number for n include more and more of the higher frequencies and therefore render a closer approximation of the original form.
   b. Epicycles have a long history as a model to explain planetary motion. Today, the concept of linked rotations is at the core of robotic motion planning, and Fourier descriptors (another term for Epicycles) are used as feature extractors for shape recognition using machine learning [38, 39].
   c. Empty, Trigonometry, Approximation

7. **Fourier2D**: The form is the result of converting the signal (binary raster image) first from the space domain to the frequency domain using Fast Fourier Transformation FFT [40], applying a low-pass filter (masking high frequencies), transforming it back to the space domain, and converting it to a binary image using a threshold.
   a. The parameter *p* adjusts the size of the square low-pass filter. Small values for *p* render a coarse approximate blob. High values for *p* include higher frequencies and hence a more accurate representation.
   b. In addition to its relevance in almost all branches of engineering and technology (signal processing, compression, dimensionality reduction), FFT presents an analogy to an architectural design process, going from an overall big gesture (low-frequency sine waves) to ever more details and smaller elements (high-frequency sines).
   c. Field, Approximation, Trigonometry, Fluid Topology
8. **Hyperbolic**: The corner points of the form are projected on a hyperbolic paraboloid. For the edges to "look" straight, they are connected with outwards bulging lines. The shortest connections between points would curve inwards and meet at less than 90°.
   a. The surface is rotated around the x-axis to reveal the three-dimensionality.
   b. Hyperboloid or other anticlastic surfaces (curvatures at any point have opposite signs) are broadly used in architecture for membrane roofs made of textile. As mathematical models, hyperbolic and spherical spaces are the two non-Euclidean geometries where the parallel postulate does not hold true.
   c. Empty, 3D
9. **LLM**: After verifying its familiarity with scalable vector graphics (SVG), a large language model (LLM, in this case ChatGPT v3.5) was prompted to generate SVG code for *"a salmon-colored rectangle of 300 by 200, rotated by 20° on a sky-blue background, canvas size of 500 × 500"* – which it provided after some initial misunderstandings.
   a. n.a.
   b. We are only at the beginning of grasping the future relevance of LLMs and other generative A.I. systems for architecture and design. Unlike for natural language, there is significantly less training data available for e.g. CAD or BIM file formats, and the physical meaning of a sequence of symbols is way more complex than a linear fairy tale. Combinations of natural language models with symbolic deduction engines like *AlphaGeometry* [41], achieving human gold-medalist accuracy at math Olympics questions are a promising solution.
   c. Empty, Uncertainty
10. **NURBS**: The form is represented as in *Point* by four corners with specific coordinates. In non-uniform rational base splines (NURBS), the points can have different degrees, controlling the shape of the curve interpolated in between. If all points have degree 1, the form is a polyline with straight edges.
    a. The parameter modifies the degree of the vertices between 1 and 5.
    b. NURBS, a generalized form of Bézier-Splines invented for the car industry in the 1950s, was central for the Blobitecture of the 2000s, the curved shapes of ZHA and liked by many for their organic appearance. They have – for better or worse – become the figurehead of computational, parametric, digital architecture.

c. Empty, Typology

11. **Offset**: The form is considered a thick line, like a brush stroke, resulting from offsetting its medial axis (the spine) to either side by half the final width.
    a. The parameter moves the positions of two of the control points on the spine polyline, thereby turning the form from straight to zig-zag.
    b. Offsetting is not only a fundamental function in design (basically a form with a material thickness) but also a widespread architectural typology. The flexibility of the spine adapts to odd-shaped plot outlines, while the constant thickness guarantees sufficient daylight and ventilation.
    c. Empty, Typology
12. **Parallelogram**: The shape is the Boolean intersection of two infinite strips of constant widths, resulting in a parallelogram. The base form results when the strips are perpendicular to each other.
    a. Rotating one of the strips maintains opposite sides to remain parallel.
    b. Parallelograms are ubiquitous in architecture, urban planning and design, more common probably than irregular polygons. They result whenever two linear elements of constant width – beams, walls, streets – intersect.
    c. Empty, Typology
13. **Point**: This is the most basic, most generic, least constrained language of all. Four points with specific coordinates $A = (x_A,y_A)$, $B = (x_B,y_B)$,… are positioned on a Cartesian coordinate system, connected with four edges $a = (A,B)$, $b = (B,C)$,… to a chain enclosing a quadrilateral face $f = (a,b,c,d)$. It is the language of drawing with a ruler or tightening a string around nails.
    a. The variable parameter is the y coordinate of one of the points. The topology (points in edges and sequence of edges) is fixed and self-intersection of edges is not prevented.
    b. The point as the minimal 2- or 3-dimensional unit with no extension is the basis for all BRep modelers. Point- (or Vertex-) coordinates together with topology information on their connectivity, which can become arbitrarily complex [42, 43], is very versatile to represent arbitrary polygons such as plots of land that developed organically as a result of topography, street networks, inheritance-induced subdivision etc. Many common exchange formats only store this information and maintain no logic on e.g. angle, distance, or parallel constraints.
    c. Empty, Typology
14. **Pointcloud**: The point cloud consists of arbitrarily many randomly distributed and unsorted points on the contour of the form. A closed contour (or surface in the 3d case) is reconstructed only afterward to best fit the points.
    a. The parameter controls the dislocation factor of the points from the original form to simulate possible noise during retrieval.
    b. Point clouds are the most basic output format of 3d scanning, a digitization method gaining importance and popularity with the increasing number of restoration and renovation projects in existing buildings. However, point clouds are not easy to work with. In their original state, they are flexible enough to contain any arbitrary distribution and, due to size, are bulky in terms of storage and computation intensity. They are often replaced with a series of best-fitting primitives, akin to new constructions (thereby losing all irregularity, patina or deformation time and use

has added to the building components. The reality lays somewhere between, used building components are neither as regular and clean as industrial products, nor as random as the cloud format allows for. Instead, they have a certain degree of stochasticity, and maybe new exchange formats containing both the genotypical information of the ideal raw type, and the phenotypical local deformations.

c. Empty, Uncertainty, Fluid Topology

15. **Polar**: The form is also described as a quadrilateral with four corner points (like Point), but instead of x and y coordinates in a Cartesian coordinate system, they are encoded with radius r and angle α in a polar coordinate system.
    a. The parameter modifies r (distance from the origin) of one of the points, thereby affecting both the visible x and y position, but not the angle α.
    b. While switching to polar coordinates in a CAD environment would be easy to implement and very helpful for the planning of circular or arc-shaped buildings, they are relatively few. Polar coordinates are nevertheless important for planetary motion, rotational mechanics and the mathematics of complex numbers to name but a few.
    c. Empty, Trigonometry
16. **Probability**: The form is a binary raster image, where each pixel's state (0/1) depends on a virtual coin flip to simulate noise in the signal. The coin's bias depends on the pixel's distance from the form boundary.
    a. The parameter $p = [0,1]$ controls the level of noise affecting the binary conversion, where $p = 0$ means no noise (original form), $p = 0.5$ means maximum noise, complete randomness, and $p = 1$ means inverted binarization, resulting in the inverse form.
    b. Vector graphics, the dominant representation in design modeling, are often praised as being superior to raster graphics for their unlimited scalability. However, their strict hard-edge binary boundary delimiting objects fail to account for continuous gradual differences or ambiguous states. An analogy can be drawn to a design process, where space attribution is initially vague, getting sharper, more distinct, and gaining confidence over time, like how the material distribution in a topology optimization process evolves and materializes from uniform abeyance over time.
    c. Field, Approximation, Uncertainty
17. **Projection**: The form results from a central projection from point P of the four corner points $ABCD$ on a plane underneath, parallel to the projected plane, resulting in the quadrilateral $A'B'C'D'$.
    a. P is moved along a circle parallel to the two planes, thereby distorting the quad. The topology is preserved, but lengths, angles, and parallelism change.
    b. From the ascent of perspectival drawing in the Renaissance, curved beams and asymmetric column distribution to counteract viewpoint distortion in Greek temples and façade elevations to legal issues with shadows cast on neighboring plots by high rises, projections are ubiquitous in architectural design and the bedrock of geometry.
    c. Empty, 3D

18. **Sphere**: The form is projected on a sphere and rendered in parallel projection. The corner points are not connected with great circles (the shortest distance between two points in spherical geometry) but with curved lines that meet at the corners at interior angles bigger than 90°.
    a. The sphere is rotated around the x-axis to reveal the three-dimensionality. With the info mode turned on, a dynamic Mercator projection of the sphere is shown as a point overlay.
    b. Spherical is a non-Euclidean geometry. As the planet we live on is close enough to a sphere and therefore theoretically impossible to lay out flat, it is of tremendous economic and political relevance, what projection method, basically what mode of distortion is chosen. Mercator projection for example lets countries further away from the equator appear gigantic [44].
    c. Empty, 3D, Trigonometry
19. **Superellipse**: This is an example of a class of purely mathematical descriptions, where the perceived form is the limit that the function (abcd) approaches asymptotically as the exponent $n \to \infty$, but will never reach.
    a. The modified parameter is the exponent n, where small numbers make the form bend inwards towards a cross shape, medium numbers result in ellipsoids of different roundness and has n gets very high, the "corners" of the form become sharper.
    b. Superellipses, related but not identical to Squircles, are primarily theoretical constructs with few practical applications. However, there are two applications – a conference table and a public circulation space (avoiding the terms "roundabout" and "square") – where a superellipse has the correct and continuously adjustable level of roundness as it blurs the binary distinction between a rectangle and an ellipse.
    c. Empty, Approximation
20. **Tessellation**: The form is the total area of the plane closer to its centroid than to any other cell. This form of tessellation of the plane is also known as Voronoi diagram. It is a generic version of tiling (all 3 regular tilings can be created with a Voronoi tessellation) as points (cell centroids) can be moved, removed or added and the topology adapts.
    a. The parameter moves the centroid of the central cell along the x-axis. Furthermore, clicking on the canvas adds additional cell centers.
    b. Tessellations, especially Voronoi, seem to have an irresistible appeal to architects, primarily for their formal and aesthetic appearance. It is essential to distinguish what looks like a Voronoi tessellation (giraffe skin, cracked porcelain, leaf veins) from what really is (closest distance cell logic). They are also relevant in data science, unsupervised ML (k-means), on graphs, and can be computed with different distance metrics. A cell's form is defined by the positions of its surrounding cells.
    c. Field, Fluid Topology, 3D
21. **Tile**: The form is one discrete piece in a regular periodic tiling of the plane with identical elements.
    a. The parameter switches between tiling with triangular, quadrangular, pentagonal and hexagonal elements.

b. Covering large surfaces by aligning small elements in a grid is ubiquitous in architecture at all scales, from thumbnail-sized mosaic stones to one-story high façade panels. Tiling is also an active field of research in geometry and has recently been extended by the discovery of an aperiodic monotile, a so-called "Einstein" [45].
c. Empty, Typology, Approximation

22. **Turtle**: The form is drawn by a little autonomous bot (virtual or physical) following a set of instructions containing of drive (for a specific time or distance) and turn (by a specific angle) commands.
a. The parameter modifies the angle of the first turn command. With the other commands remaining unchanged, the form is no longer closed.
b. Turtle robots have been a famous class of educational devices since their invention in the 1940s, and they flourish in many versions today. Small autonomous robots with a local coordinate system and the orchestration and control of multiple of them are also increasingly important on construction sites.
c. Empty, Typology

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